

KOCHI METRO RAIL LIMITED (KMRL)

Maintenance Division - Track Inspection Report

Date: August 15, 2026

Report ID: KMRL-MNT-2026-0847

Location: Aluva - Palarivattom Corridor, Track Section 4B

Inspector: Rajesh Kumar, Senior Engineer

EXECUTIVE SUMMARY

This report documents the findings of the routine track inspection conducted on August 14, 2026, along the Aluva to Palarivattom corridor. The inspection covered a total distance of 8.5 kilometers and focused on rail integrity, sleeper conditions, ballast stability, and overhead electric line (OHE) alignment.

KEY FINDINGS

- Rail Condition:** Minor surface corrosion detected on rails between chainage 2450m and 2670m. Corrosion depth measured at 0.8mm, within acceptable limits but requires monitoring.
- Sleeper Inspection:** Three concrete sleepers (IDs: SL-4421, SL-4422, SL-4423) show hairline cracks. Structural integrity remains intact. Recommended for replacement within 90 days.
- Ballast Analysis:** Ballast depth adequate (average 300mm). However, drainage channels near chainage 3100m are partially blocked by debris. Cleaning required to prevent waterlogging during monsoon.
- OHE Alignment:** Overhead wire tension within specified range (15.5 kN). Pantograph contact smooth. One insulator (INS-889) shows early signs of carbon tracking. Replacement scheduled.
- Signaling Equipment:** Track circuits functioning normally. Axle counters calibrated and verified. No anomalies detected in signal interlocking systems.

RECOMMENDATIONS

Priority 1: Clear drainage channels within 7 days before forecasted heavy rainfall.

Priority 2: Replace cracked sleepers and faulty insulator during next maintenance window (August 22-23, 2026).

Priority 3: Apply protective coating to corroded rail section during non-operational hours.

SAFETY OBSERVATIONS

All safety protocols were followed during inspection. No unauthorized personnel entered the track area. Emergency communication systems tested and confirmed operational.

The overall track condition is rated as GOOD with minor maintenance requirements. No speed restrictions recommended at this time.

Approved by:

Arun Nair

Chief Maintenance Engineer

KMRL Operations

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APPENDIX A - DETAILED MEASUREMENTS

Rail Gauge Measurements (mm):

Chainage 2400m: 1435mm (Standard)

Chainage 2500m: 1434mm (Normal)

Chainage 2600m: 1436mm (Normal)

Chainage 2700m: 1435mm (Standard)

Chainage 2800m: 1433mm (Requires Adjustment)

Sleeper Spacing (mm):

Average spacing: 600mm

Maximum deviation: +12mm at chainage 2550m

Minimum deviation: -8mm at chainage 2780m

Ballast Gradation:

Passing 63mm sieve: 100%

Passing 50mm sieve: 95%

Passing 40mm sieve: 65%

Passing 20mm sieve: 15%

All measurements comply with Indian Railways standards IRC:24-2010 and KMRL internal specifications KMRL-ENG-2019-v3.

Document prepared in accordance with ISO 9001:2015 quality management standards.